

Study Guide: Professional Skills and Behaviours for Cyber Security (Unit 6)

1. Digital Identity and Screening Checks

1.1 The importance of managing and promoting a positive digital identity

- **Digital footprint:** The traces you leave online (social media, blogs, posts, mentions).
- **Positive digital identity:** Showcases professionalism and technical awareness.
- **Why it matters:** Employers check online presence; no online footprint may imply lack of tech knowledge or hiding something.

Tip: Maintain LinkedIn, professional portfolios, and clean social media profiles.

1.2 Possible employee screening checks

UK employers can legally perform:

1. **Right to work checks** – confirm eligibility to work in the UK.
 2. **Criminal record checks (DBS checks)** – ensure suitability, especially in sensitive roles.
 3. **Credit checks** – for financial or law-related positions.
- Other checks include: references, medical checks, and social media screening.

1.3 Potential consequences of unsatisfactory findings

- Failing checks may lead to:
 - Job offer withdrawal
 - Reputation damage
 - Ineligibility for government roles

Key point: Pre-employment checks protect both employer and organisation from risk.

1.4 Levels of security clearance

Level	Purpose	Notes
BPSS	Standard background checks for identity, nationality, employment history	Not formal clearance
SC (Security Check)	Access to Secret assets	Includes criminal, financial, MI5 checks
CTC (Counter Terrorist Check)	Roles near public figures or sensitive locations	Targeted for security/terrorism risk
DV (Developed Vetting)	Top Secret access	Very detailed: finances, interviews, references

1.5 How bias can influence cyber security

- **Bias:** Prejudice for/against a person or group.
- **Confirmation bias:** Paying attention only to evidence that supports what you already think.
- **Other biases:**
 - **Fundamental attribution error / PEBKAC:** Blaming user for security incident without considering system faults.
 - **Aggregate bias:** Assuming an individual behaves like their group.
 - **Framing effect:** Influence of how information is presented.

Impact: Can lead to wrong decisions, overlooked threats, or misdirected investigation.

2. Skills and Personal Development

2.1 Skills required for a career in cyber security

- **Technical skills:** Networking, malware detection, incident response.
- **Soft skills:** Communication, attention to detail, problem-solving, continuous learning.

2.2 Personal skills analysis

- Use **SWOT** to self-assess:
 - **Strengths** – What you're good at
 - **Weaknesses** – Areas for improvement
 - **Opportunities** – Career options, trends
 - **Threats** – Obstacles or competition
- Alternative: **SOAR analysis** (focuses on Strengths, Opportunities, Aspirations, Results) for growth-oriented planning.

2.3 Assess skills against career requirements

- Compare your technical and soft skills with job/industry requirements.
- Identify **gaps** and areas to improve.

2.4 Create a personal development plan (PDP)

- **SMART Goals:**
 - **Specific**
 - **Measurable**
 - **Achievable**
 - **Relevant**
 - **Time-bound**
- Helps track skills growth and readiness for roles in cyber security.

3. Continuing Professional Development (CPD)

3.1 What is CPD?

- Ongoing process to **maintain and develop knowledge and skills** in your career.
- Essential in cyber security due to constant new threats and technologies.

3.2 Methods to keep up to date

- **Reading industry news / blogs** (e.g., cyber security publications)
- **Attending training courses / workshops**
- **Professional networking** (LinkedIn, forums, conferences)
- **Certifications** (e.g., CompTIA Security+, CEH, CISSP)

3.3 Why CPD is important

- Protects your organisation from new threats
- Keeps your knowledge current
- Increases employability and career growth

Glossary of Key Terms

Term	Definition
Aggregate Bias	When people make assumptions about an individual using information that describes the broader group the individual is associated with.
Baseline Personnel Security Standard (BPSS)	The standard pre-employment security screening. It is not a formal security clearance but is required for many government-related jobs.
Bias	A preference or prejudice for or against one person or group, especially in a way considered to be unfair.
Confirmation Bias	When people pay more attention to, or give more weight to, information that supports what they already think or believe.
Continuing Professional Development (CPD)	The process of maintaining and developing the skills, knowledge and experience that you need in your work or profession.
Counter Terrorist Check (CTC)	A security clearance required for government employees and contractors whose roles involve proximity to public figures, access to material valuable to terrorists, or unescorted access to at-risk establishments.

Critical Thinking	The deliberate and systematic processing of information to help make better decisions, involving analysis, evaluation, and prioritization of information.
Developed Vetting (DV)	The highest level of security clearance, used for posts which require frequent and uncontrolled access to Top Secret assets or code word material.
Digital Footprint	The collection of traces left by an individual's online activity, which may include social media posts, blogs, published items, and mentions in search results.
Framing Effect	A type of bias that occurs when information is presented in a way that emphasizes a particular perspective, often using aggressive or negative wording to influence decisions.
Fundamental Attribution Error	The tendency to blame a user for a security incident without looking for other types of causes. Also known as PEBKAC.
PEBKAC	An acronym for 'Problem Exists Between Keyboard and Chair', another term for Fundamental Attribution Error, where the user is blamed for a security incident.
Personal Development Plan	A roadmap for acquiring the skills needed to achieve a specific goal, often created after a personal skills analysis.
Pre-employment Checks (Vetting)	Checks carried out by human resources teams to ensure a candidate will not bring the company into disrepute or cause other difficulties. They are a key part of due diligence.
Security Check (SC)	A security clearance required for people who will have long-term, frequent, and uncontrolled access to Secret assets and/or occasional, supervised access to Top Secret classified assets.
Security-by-Design	A mindset or philosophy where a system or product has been designed from the ground up to emphasize security, rather than adding security features onto an existing system.
SMART Goals	A framework for setting goals that are Specific, Measurable, Achievable, Relevant, and Time-bound.
SOAR Analysis	A personal skills analysis tool that focuses on S trengths, O pportunities, A spirations, and R esults to build a strengths-based strategy.
SWOT Analysis	A self-assessment method used to identify internal S trengths and W eaknesses, and external O pportunities and T hreats.